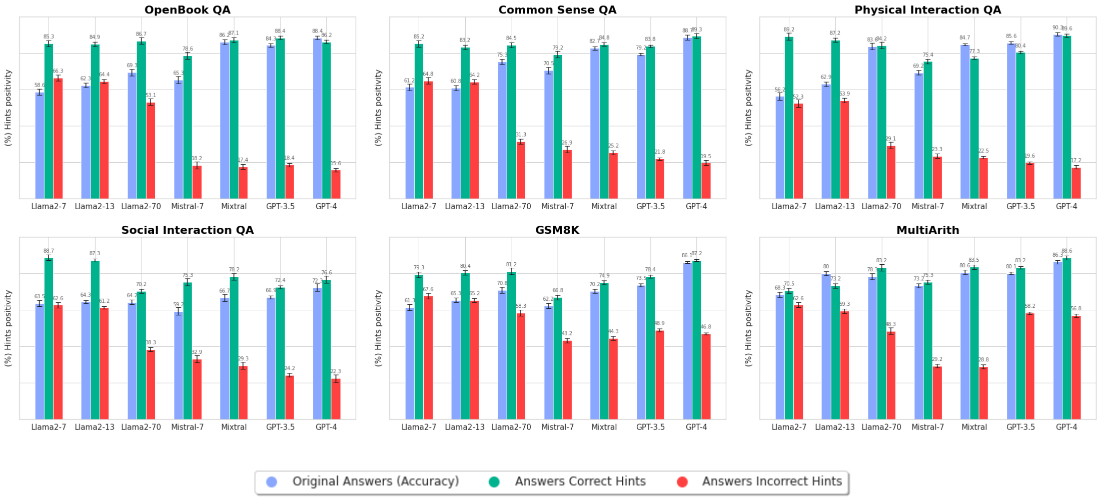

Figure 6: We examine the Self-confidence and sycophantic behaviours of LLMs in §2.3 in question-answering tasks. We use subsets of four datasets (§3.3). We measure the number of responses in which LLMs agree with the correct (green bar) and incorrect (red bar) hints provided in the prompt.

results in Figure 4 demonstrate that the percentage of non-contradiction is higher when the entities are poets; instead, it is lower when the entities are public persons (experimentation setting detailed in Appendix F). Finally, to complete the analysis in Appendix J, we propose an additional evaluation showing that this phenomenon occurs beyond poetry tasks as well. Specifically, we observed comparable results by proposing a task to summarise biographies belonging to the wrong people. This result shows that the problem is not limited to the poetry task.

In conclusion, we show that although the models examined tend to satisfy the users' requests and viewpoints by providing satisfactory answers to the speaker and often without emphasizing possible errors in the prompt, the need remains to examine limited contexts such as those present in question-answering benchmarks. To study this latter aspect, we continue the analysis in §4.3

4.3 Behind Self-confidence lies Robustness?

LLMs seem Self-confident in their choices, particularly in question-answering tasks where there is little space for users' point-of-view and perspectives. However, there are some exceptions, as in Figure 1, where GPT-3.5 disagrees with hints in prompts when they are incorrect, but this is not always true, as Llama2-70 and Mixtral seem to follow the misleading hint.

Self-confidence & Performances The best-performing LLMs, i.e., those with higher accuracy values (blue bars in Figure 6), appear not to follow users' misleading hints and significantly improve performance when the hints are correct. This phenomenon is present mainly in the four multiple-choices question-answering tasks on Llama-2-70, Mixtral, and both GPTs (first four plots in Figure 6), while the gap is not in the same scale in the mathematical tasks (GSM8K and MultiArith in Figure 6). On the other hand, the models with fewer parameters (Llama2-7 and Mistral-7) not only have a lower baseline performance than the other models but also seem to follow the users' hints on all task types to a greater scope (red bars in Figure 6). As observed, the self-confidence rate does not always have the same performance. Therefore, we heighten the analysis by exemplifying the hints in the different task types and performing a more accurate analysis of the motivations behind following bad hints.

The Hints Role LLMs are sensitive to misleading hints. The results obtained in GSM8K and MultiArith show high agreement rates in misleading hints (red bars Figure 6). Therefore, we repeat the experiments, proposing different kinds of prompts.

Hence, as described and discussed in Appendix H, highly misleading hints do not seem to have the previously discussed effects. Instead, it appears that LLMs produce radically different outputs that contradict the hints provided by users, as shown in Figure 5. Merely as happened in the experiments in §3.1, prompts containing misleading information but very close to the target domain (e.g., in the Non-contradiction task, poets close to the actual writer, and in these tasks, numbers very close to the target numbers) raise the bar and the generalisation challenges of the LLMs by promoting causal generations that tend to meet and mimic the prompt.

Self-confidence vs Parameters However, the models with fewer parameters underperform those with more parameters, both in benchmarking with the original prompt and versions with correct and incorrect hints. However, we reproduce the experiments on a limited subset to observe the behaviours in instances that are generally classified correctly, as described in Appendix G. Figure 8 (discussed in Appendix G) shows that the agreement rates with incorrect hints drop significantly when prompts are altered for which the models generate the correct answer. This result indicates that, although the lower-performing LLMs have been shown to follow prompts with mistakes in the overall experiments, the underlying motivations could be related to poor performances on original tasks on particular subsets of instances.

5 Related Works

Although human feedback has proven to be an excellent component for refining the interaction between the user and the Large Language Model (LLM), this method can bring some adverse effects, such as sycophancy. In particular, it seems that the mechanism of Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2023) stimulates the model to consider the users' opinion (Perez et al., 2022) as the right truth. However, if the model prefers the users' opinion over the correct answer, regardless of whether it is correct, it brings robustness and reliability issues. The initial attitudes related to the input prompt were highlighted by Zhao et al. (2021), who emphasised the propensity of LLMs to provide responses related to the input or commonly present in the pre-training dataset. Building upon this statement, Lu et al. (2022) demonstrated how the specific arrangement of examples can vary the models' performance from state-of-the-art to random-guessing performance. Similarly, Turpin et al. (2023) discovered that in a chain-of-thought context (Wei et al., 2023a), language models can be easily influenced toward specific responses. Moreover, they showed the presence of high bias factors due to prompt sensitivity. The sensitivity of the prompt and interactions with users seem to be pivotal points in the study of LLMs' resilience. Perez et al. (2022), by introducing the concept of sycophancy, showed the behaviours of these models not to contradict human ideas and points of view embedded in the prompt. Wei et al. (2023b) proposed a data-level intervention to avoid LLMs' sycophantic behaviours. Finally, Sharma et al. (2023) adopted the experiments proposed by Wei et al. (2023b) by extending the models under investigation and proposing further data to understand the weight of RLHF in sycophantic behaviours.

In this paper, we propose a comprehensive analysis of the attitudes of LLMs. Building on previous work (summarised in Table 1), we introduce systematic interventions that influence prompts with misleading hints and opposite points of view. Our contributions are as follows: (i) We discuss different tasks where we probe and analyse the sycophantic behaviour of LLMs via a systematic series of interventions. Hence, starting from existing resources, we extend them by instilling human-influenced beliefs and real or misleading hints. (ii) We provide a robust analysis of different LLMs by examining the impact of human feedback-based refinement techniques on their behaviour. (iii) We show that LLMs tend to follow the views the user expresses, except when the response choice is strict.

6 Conclusion

This paper highlights a critical aspect of Large Language Models (LLMs) and their suggestibility to sycophantic behaviour. While LLMs have shown outstanding abilities in solving complex tasks and aligning with human evaluations, this adaptability also introduces a tendency to generate responses that may align more with users' beliefs rather than factual accuracy. We discern scenarios that could induce LLMs to have sycophantic behaviour by proposing different interventions for several tasks. From downstream results, it emerges that LLMs exhibit sycophantic behaviour and agree with user beliefs, especially in contexts involving subjective opinions or when factual contradictions are expected. At the same time, these attitudes are significantly less pronounced in objective decision-making scenarios. However, the overall analysis highlights the partial robustness of these models and raises critical weaknesses in reliability in critical decision-making scenarios.

7 Limitations & Future Works

In this work, we studied the tendencies of LLMs to produce responses in line with users even in the presence of errors or mistakes, a behaviour known as sycophancy. In particular, we analysed this on question-answering benchmarks and observed that the models of the GPT family are very robust and do not get influenced by human-influenced prompts. Although this seemed animating from a stability point of view, it was not confirmed in further analyses. In fact, by systematically asking for opinions in contexts strongly guided by human opinion, the GPTs also did not counter the latter. Finally, we tested the tendency to mimic human errors even in the presence of obvious mistakes. Similarly, the Llama family models and the GPTs showed minor disagreement with prompts specially manipulated to simulate human error.

Even though our experiments stably show sycophantic tendencies of LLMs to follow prompt content, there are limits to be considered. First, the behaviour we describe as sycophantic on LLMs we have observed principally in two models with fewer parameters, namely those of the Llama family. This does not demonstrate that. Indeed, the LLMs' attitudes are due to human feedback refinement techniques (as hypothesized in (Sharma et al., 2023)). Our analysis is limited to empirically describing the response rate following feedback influenced by synthetically constructed prompts inspired by human behaviour. We intend to provide further analysis and strengthen our current methods in future developments. Firstly, we would like to epistemically understand if there are relationships between the topics of human-influenced prompts where LLMs agreed and those where they disagreed. Secondly, we plan to expand our analyses by correlating the impact of human feedback with the obtained results. Thirdly, we intend to produce additional resolutions to help understand human errors and interactions with LLMs. Fourthly and finally, we would like to extend our models to additional well-known LLMs.

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